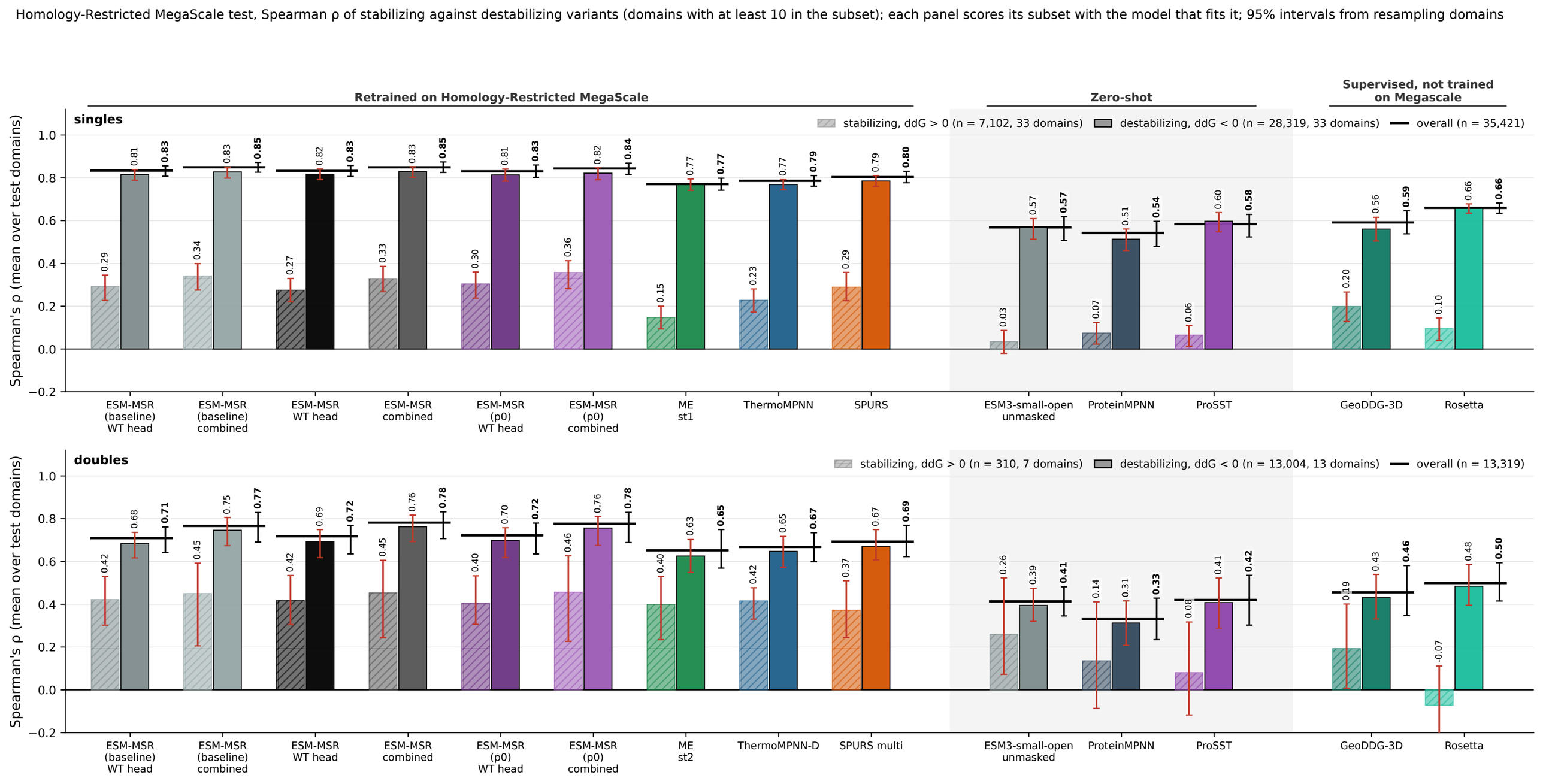
Homology-Restricted MegaScale test, Spearman ρ of stabilizing against destabilizing variants (domains with at least 10 in the subset); each panel scores its subset with the model that fits it; 95% intervals from resampling domains
Retrained on Homology-Restricted MegaScale
Zero-shot
Supervised, not trained on Megascale
singles
stabilizing, ddG > 0 (n = 7,102, 33 domains)
destabilizing, ddG < 0 (n = 28,319, 33 domains)
overall (n = 35,421)
Spearman's ρ (mean over test domains)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
0.29 0.81 0.83
0.34 0.83 0.85
0.27 0.82 0.83
0.33 0.83 0.85
0.30 0.81 0.83
0.36 0.82 0.84
0.15 0.77 0.77
0.23 0.77 0.79
0.29 0.79 0.80
0.03 0.57 0.57
0.07 0.51 0.54
0.06 0.60 0.58
0.20 0.56 0.59
0.10 0.66 0.66
ESM-MSR (baseline) WT head
ESM-MSR (baseline) combined
ESM-MSR WT head
ESM-MSR combined
ESM-MSR (p0) WT head
ESM-MSR (p0) combined
ME st1
ThermoMPNN
SPURS
ESM3-small-open unmasked
ProteinMPNN
ProSST
GeoDDG-3D
Rosetta
doubles
stabilizing, ddG > 0 (n = 310, 7 domains)
destabilizing, ddG < 0 (n = 13,004, 13 domains)
overall (n = 13,319)
Spearman's ρ (mean over test domains)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
0.42 0.68 0.71
0.45 0.75 0.77
0.42 0.69 0.72
0.45 0.76 0.78
0.40 0.70 0.72
0.46 0.76 0.78
0.40 0.63 0.65
0.42 0.65 0.67
0.37 0.67 0.69
0.26 0.39 0.41
0.14 0.31 0.33
0.08 0.41 0.42
0.19 0.43 0.46
-0.07 0.48 0.50
ESM-MSR (baseline) WT head
ESM-MSR (baseline) combined
ESM-MSR WT head
ESM-MSR combined
ESM-MSR (p0) WT head
ESM-MSR (p0) combined
ME st2
ThermoMPNN-D
SPURS multi
ESM3-small-open unmasked
ProteinMPNN
ProSST
GeoDDG-3D
Rosetta